



03 — Replay Engine & Prototype Formation

Overview

The replay engine is the **hippocampal replay** analogue: periodically, it samples a batch of episodic memories, assigns each to the nearest semantic prototype (or creates new ones), and updates the prototype graph edges. This is the bridge from raw episodic traces to stable semantic prototypes.

Mathematical Formulation

Step 1: Saliency-Proportional Sampling

Sample N episodes without replacement:

$$\mathcal{B} = \{e_i \sim P(i) \mid i = 1 \dots N\}, \quad P(i) = \frac{s_i}{\sum_j s_j}$$

Where $N = \text{sample_size}$ (default: 256).

Step 2: Prototype Assignment (Dual-Scale)

For each sampled episode e with embedding $\mathbf{e}$:

For $\text{scale} \in \{\text{fine}, \text{coarse}\}$:

$$\mathcal{P}_{\text{scale}} = \text{FAISS.search}(\mathbf{e}, \text{top_k} = 1, \text{scale})$$

Let p^* be the nearest prototype with centroid $\mathbf{c}^*$, and let $\cos^* = \langle \mathbf{e}, \mathbf{c}^* \rangle$.

Dentate Gyrus gate (Stage 8, optional):

When `use_pattern_separation = True`, find the runner-up prototype p' with similarity $\cos'$:

$$\text{distinctive_sim} = \cos^* - \lambda_{\text{sep}} \cdot \cos'$$

Where $\lambda_{\text{sep}} = \text{dg_separation_lambda}$ (default: `0.3`).

If `distinctive_sim < assignment_threshold(scale)`, the episode creates a **new** prototype.

Assignment threshold per scale:

Scale	Default Threshold	Brain Analogue
<code>fine</code>	<code>assignment_threshold</code> (0.85)	CA3: high specificity
<code>coarse</code>	<code>coarse_assignment_threshold</code> (0.60)	CA1: broad clusters

Step 3: Prototype Update (Online Mean)

When episode e is assigned to prototype p :

$$\mathbf{c}_{\text{new}} = \frac{n \cdot \mathbf{c}_{\text{old}} + \mathbf{e}}{n + 1}$$

$$n_{\text{new}} = n_{\text{old}} + 1$$

$$\sigma_{\text{new}}^2 = \frac{n - 1}{n} \cdot \sigma_{\text{old}}^2 + \frac{\|\mathbf{e} - \mathbf{c}_{\text{new}}\|^2}{n}$$

For all prototypes p, q co-occurring in the replay batch:

Similarity weight:

$$w_{\text{sim}}(p, q) = \cos(\mathbf{c}_p, \mathbf{c}_q)$$

Transition weight (EMA):

$$w_{\text{trans}}(p, q) \leftarrow w_{\text{trans}}(p, q) \cdot \alpha_d + \text{count}(p \rightarrow q) \cdot (1 - \alpha_d)$$

Coactivation weight (EMA):

$$w_{\text{coact}}(p, q) \leftarrow w_{\text{coact}}(p, q) \cdot \alpha_d + \text{count}(p, q) \cdot (1 - \alpha_d)$$

Where $\alpha_d = \text{accumulate_decay}$ (default: `0.3`).

Fused edge weight:

$$w(p, q) = \lambda_1 \cdot w_{\text{sim}}(p, q) + \lambda_2 \cdot w_{\text{trans}}(p, q) + \lambda_3 \cdot w_{\text{coact}}(p, q)$$

Parameter	Default	Weight
<code>lambda_similarity</code> λ_1	<code>1.0</code>	Cosine similarity between centroids
<code>lambda_transition</code> λ_2	<code>0.5</code>	Temporal succession probability
<code>lambda_coactivation</code> λ_3	<code>0.3</code>	Co-occurrence in same batch

Step 5: Homeostatic Normalization

Per-source L1 normalization:

$$w(p, q) \leftarrow w(p, q) \cdot \frac{T}{\sum_{q'} w(p, q')}, \quad T = \text{homeostatic_target}$$

Relative pruning:

$$\text{prune } w(p, q) \text{ if } w(p, q) < \max(\tau_{\text{ratio}} \cdot \max_{q'} w(p, q'), \tau_{\text{abs}})$$

Where:

- $T = \text{homeostatic_target}$ (default: `0.5`)
- $\tau_{\text{ratio}} = \text{prune_ratio}$ (default: `0.2`)
- $\tau_{\text{abs}} = \text{prune_below}$ (default: `0.05`)

Step 6: Self-Supervised Rehearsal (Stage 5)

For selected prototypes (most recent member as probe):

1. Retrieve top-k episodes for probe embedding
2. **Misses:** siblings from other prototypes not retrieved → reinforce coactivation bridge (+ `self_supervise_miss_reward`)
3. **Confusers:** foreign prototype episodes in results → penalize coactivation bridge (− `self_supervise_confuser_penalty`)
4. Skip prototypes with `needs_review` or `contradicting_episode_ids` schemas (dentate gyrus gate)

Configuration

ReplayConfig

Parameter	Type	Default	Description
sample_size	int	256	Episodes sampled per replay pass
max_prototypes_per_replay	int	32	Max new prototypes per pass
assignment_threshold	float	0.85	Cosine threshold for CA3 (fine)
coarse_assignment_threshold	float	0.60	Cosine threshold for CA1 (coarse)
transition_batch_size	int	64	Batch for transition counting
transition_steps	int	50	Max temporal steps for transitions
use_pattern_separation	bool	False	Dentate gyrus pattern separation
dg_separation_lambda	float	0.3	Runner-up penalty strength
self_supervise	bool	True	Self-supervised rehearsal
self_supervise_max_prototypes	int	32	Max prototypes probed
self_supervise_min_members	int	3	Min members to probe
self_supervise_top_k	int	8	Top-k retrieval for probe
self_supervise_miss_reward	float	0.5	Coactivation boost for misses
self_supervise_confuser_penalty	float	0.25	Coactivation penalty for

Parameter	Type	Default	Description
			confusers

Key Invariants

- 1. Prototype centroids are online means — converge over multiple passes.
- 2. Edge weights EMA with `accumulate_decay = 0.3` — 70% current pass, 30% history.
- 3. Homeostatic normalization prevents super-hub dominance.
- 4. Self-supervised rehearsal is failure-driven: only misses/confusers produce signals.
- 5. Contradicted/review schemas are excluded from rehearsal probes (dentate gyrus gate).